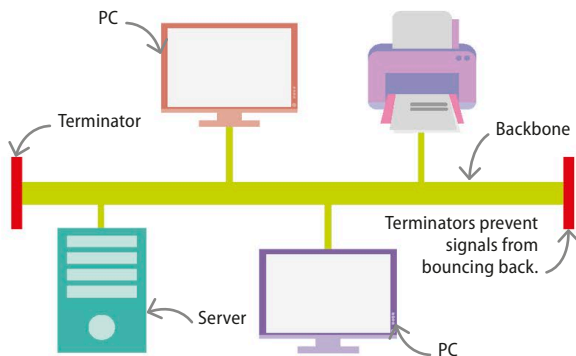


Topology

The topology of a network – also known as its layout – is the strategy used to connect devices together. The layout depends on the type of information being shared, the volume of communications, and how devices need to store data. There are pros and cons to each layout.

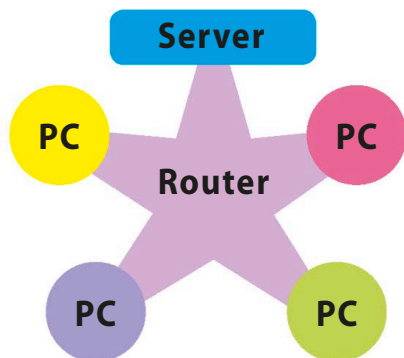
Bus topology

In this topology, devices are connected to a single main wire called the backbone, or bus. If a computer wants some data, it sends a request to all the devices in the network, but only the target device responds. Bus topology is cheap and simple to set up. However, if the backbone breaks, the network is useless. Nowadays, bus topology is considered outdated.



Star topology

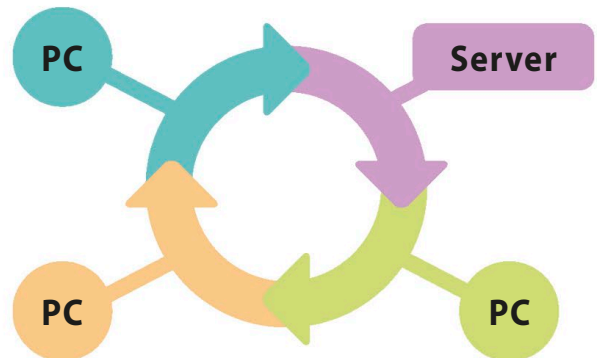
In this topology, each device is connected to a single hub, typically a router. All communications pass through this hub. Star topology can also be called octopus topology because the hub can have many connections, or tentacles. It is cheap and easy to expand and handles breaks well – unless the hub goes down, in which case the entire network is out.



Many networks are **hybrids**, created by combining **multiple topologies**.

Ring topology

In this topology, devices are connected to a central ring where data flows in a single direction. When a computer sends a request, the signal travels along the ring and visits each device. Ring topology handles volume better than bus topology. It's more robust and can span greater distances. However, it's more expensive, and like bus topology, it's considered outdated.



Mesh topology

In this topology, every node is connected to every other node. It handles breaks better than other topologies because there are many routes that data can take between any two devices. Mesh topology is very expensive and requires lots of cable. It isn't usually used for LANs, but can be a good layout for MANs or WANs.

